so that $h^{1-\gamma}\tilde{v}_h$ has the same limit as v_h, while $\min\{h^{-\gamma}, h^{2-2\gamma}\}\tilde{u}_h$ converges to some $\tilde{u}$, representing the same displacement as u expressed in a different basis. Note that the same argument would apply defining $\tilde{u}_h$ in terms of a basis of the form $\{v_1, v_2, U_j^{-1}e_3\}$, with $v_1, v_2 \in \operatorname{span}\{U_j e_1, U_j e_2\}$.

Corollary 3.11. *In the same notation and hypothesis of Proposition 3.9, there exists a map $G \in L^2(\Omega, \mathbb{R}^{3\times 3})$ such that, up to a subsequence,*

$$G_h := h^{-\gamma}\left(\tilde{R}_h^T \nabla_h \tilde{y}_h - U_j\right) \rightharpoonup G \text{ in } L^2(\Omega; \mathbb{R}^{3\times 3})\,. \tag{30}$$

Moreover, $G^{1,2}$ is affine in x_3, that is

$$G^{1,2}(x', x_3) = G_0(x') + x_3 G_1(x')\,.$$

Finally,

$$(U_j G_1)' = -\nabla^2 v$$

and

$$\begin{aligned}
&(\operatorname{sym}(U_j G_0))' = \operatorname{sym}(\nabla u) && \text{if } \alpha > 4\,,\\
&(\operatorname{sym}(U_j G_0))' = \operatorname{sym}(\nabla u) + \frac{1}{2}|U_j^{-1}e_3|^2 \nabla v \otimes \nabla v && \text{if } \alpha = 4\,,\\
&\nabla u + \nabla u^T + |U_j^{-1}e_3|^2 \nabla v \otimes \nabla v = 0 && \text{if } 2 < \alpha < 4\,.
\end{aligned} \tag{31}$$

Proof. We will write U in place of U_j to simplify the notation. Convergence (30) follows immediately from (20). To show that $G^{1,2}$ is affine, we study the difference quotient

$$H_h^s(x', x_3) = \frac{1}{s}(G_h(x', x_3 + s) - G_h(x', x_3))\,,$$

for s such that $x_3 + s \in I$. Repeating the same computation as in Corollary 3.7, we deduce that for $\alpha = 1, 2, 3$ and $\beta = 1, 2$ we have

$$(\tilde{R}_h(x') H_h^s(x', x_3))_{\alpha\beta} = \frac{1}{s}\frac{\partial}{\partial x_\beta}\int_0^s h^{1-\gamma}\left(\frac{1}{h}\frac{\partial \tilde{y}_{h,\alpha}}{\partial x_3}(x', x_3 + \sigma) - E_{\alpha 3}\right) d\sigma\,. \tag{32}$$

By (24), the integral on the right hand-side converges strongly as $h \to 0$ in $L^2(\Omega)$ to

$$\int_0^s (AU)_{\alpha 3}\, d\sigma = s(AU)_{\alpha 3}\,.$$

Hence, the right-hand side of (32) converges strongly in $(W_0^{1,2}(\Omega))^*$ to $\frac{\partial}{\partial x_\beta}[(AU)_{\alpha 3}]$. By (30), the left-hand side of (32) converges weakly in $L^2(\Omega)$ to

$$H^s(x', x_3)_{\alpha\beta} := \frac{1}{s}(G(x', x_3 + s)_{\alpha\beta} - G(x', x_3)_{\alpha\beta})\,. \tag{33}$$

Since $L^2(\Omega)$ is continuously embedded in $(W_0^{1,2}(\Omega))^*$, we obtain

$$H^s(x', x_3)_{\alpha\beta} = \frac{\partial}{\partial x_\beta}[(AU)_{\alpha 3}] \quad \forall\, \alpha = 1, 2, 3 \quad \forall\, \beta = 1, 2\,. \tag{34}$$

In particular, the first two columns of H^s are independent of x_3 (recall that A depends only on x') and so the first two columns of G are affine in x_3.

For the final part of the statement note that G_h can be rewritten as follows

$$G_h = \frac{\nabla_h \tilde{y}_h - U_j}{h^\gamma} - \frac{R_h U_j - U_j}{h^\gamma} + (R_h - \mathrm{Id})^T \frac{\nabla_h \tilde{y}_h - R_h U_j}{h^\gamma} .$$

Hence,

$$\begin{aligned}(U_j G_h)' = (U_j)_{1,2} \frac{\nabla_h \tilde{y}_h U_j^{-1} - \mathrm{Id}}{h^\gamma} (U_j)^{1,2} - h^{\gamma-2} (U_j)_{1,2} \frac{R_h - \mathrm{Id}}{h^{2\gamma-2}} (U_j)^{1,2} \\ + \left[U_j (R_h - \mathrm{Id})^T \frac{\nabla_h \tilde{y}_h - R_h U_j}{h^\gamma} \right]' .\end{aligned} \tag{35}$$

If $\gamma > 2$, then $\min\{h^{-\gamma}, h^{2-2\gamma}\} = h^{-\gamma}$. Integrating with respect to x_3, taking the symmetric part and passing to the limit as $h \to 0$ we get

$$\mathrm{sym}(U_j G_0)' = \mathrm{sym}(\nabla u) ,$$

where we have used (20), (23) and (25). If $\gamma = 2$, passing to the limit in (35) we also get the term $-\frac{(UA^2U)'}{2}$. By the characterization (26) of UAU we get with some computation

$$\begin{aligned}(UA^2U)' &= (UAU(U^{-1})^2 UAU)' = (UAU)_{1,2} (U^{-1})^2 (UAU)^{1,2} \\ &= -\nabla v^T (U^{-1} e_3)^T U^{-1} e_3 \nabla v = -|U^{-1} e_3|^2 \nabla v \otimes \nabla v ,\end{aligned}$$

proving (31). Lastly, if $1 < \gamma < 2$, we multiply both sides of (35) by $h^{2-\gamma}$ before passing to the limit. The left-hand side converges to 0, while the right-hand side converges again to

$$\mathrm{sym}(\nabla u) + \frac{1}{2} |U_j^{-1} e_3|^2 \nabla v \otimes \nabla v .$$

Finally, note that $G_1 = H^1$ by (33). Thus, by (34) for $\alpha, \beta = 1, 2$

$$(UG_1)_{\alpha\beta} = \sum_{k=1}^{3} (U)_{\alpha k} \sum_{l=1}^{3} \frac{\partial A_{kl}}{\partial x_\beta} (U)_{l3} = \frac{\partial}{\partial x_\beta} (UAU)_{\alpha 3} = -\frac{\partial^2 v}{\partial x_\beta \partial x_\alpha} .$$

■

Proof of Theorem 1.3–(i). It immediately follows from Proposition 3.9 and Corollary 3.11. ■

4 Proof of the Γ-Convergence results

We are now ready to complete the proofs of Theorem 1.1 and Theorem 1.3. By the results of the previous section, we just need to prove the lim inf inequality and the existence of recovery sequences.

4.1 The lim inf inequality

Proof of Theorem 1.1–(ii) and Theorem 1.3–(ii). Define the matrix G_h as in Corollary 3.11 or Corollary 3.7, depending on the value of α. Let $\tilde{\Omega}_h := \{x \in \Omega : |G_h(x)| < h^{-1}\}$ and let χ_h be its characteristic function. Clearly, χ_h is bounded and $\chi_h \to 1$ in $L^1(\Omega)$. Thus, we have $\chi_h G_h \rightharpoonup G$ in $L^2(\Omega)$. Expanding W, we get

$$W(\nabla_h y_h) \geq W(U_j + h^\gamma G_h) \geq \frac{1}{2} Q_j(h^\gamma G_h) - \omega(h^\gamma |G_h|) h^\alpha |G_h|^2 ,$$

where ω is the modulus of continuity of D^2W at U_j. In particular

$$\begin{aligned} E_h^\alpha(y_h) &\geq \frac{1}{2}\int_\Omega \left[Q_j(\chi_h G_h) - \omega(h^\gamma|\chi_h G_h|)|\chi_h G_h|^2\right]\,dx \\ &\geq \frac{1}{2}\int_\Omega Q_j(\chi_h G_h)\,dx - C\omega(h^{\gamma-1})\,. \end{aligned}$$

Recall that Q_j is weakly lower semicontinuous in $L^2(\Omega)$ by convexity. Thus, passing to the limit and applying Lemma 2.1 we obtain

$$\begin{aligned} \liminf_{h\to 0} E_h^\alpha(y_h) &\geq \frac{1}{2}\int_\Omega Q_j(G)\,dx = \int_\Omega Q_j(U_j^{-1}\operatorname{sym}(U_j G))\,dx \\ &\geq \int_\Omega \bar{Q}_j(\operatorname{sym}(U_j G)')\,dx\,. \end{aligned}$$

Recall that

$$\operatorname{sym}(U_j G(x',x_3))' = \operatorname{sym}(U_j G_0(x'))' + x_3\operatorname{sym}(U_j G_1(x'))'\,.$$

By Corollary 3.7 and Corollary 3.11 we conclude. ■

4.2 Recovery sequences

We are left with the construction of the recovery sequences. Here, for a clearer exposition, we follow the reverse order and start with the recovery sequence for the case $\alpha > 4$. We will often use the big(small)-O notation, that has to be understood uniformly over Ω.

Proof of Theorem 1.3–(iv). Let $j \in \{1,\dots,l\}$. Suppose $\alpha > 4$. By a standard density argument it is sufficient to exhibit a recovery sequence for $u \in C^\infty(\bar{S};\mathbb{R}^2)$ and $v \in C^\infty(\bar{S})$. We define

$$B_h(x',x_3) := \begin{pmatrix} h^\gamma u \\ h^{\gamma-1}v \end{pmatrix} - h^\gamma x_3 \begin{pmatrix} \nabla v^T \\ 0 \end{pmatrix} + \frac{h^{\gamma+1}}{2}x_3^2\xi(x') + h^{\gamma+1}x_3\zeta(x')\,,$$

and

$$y_h(x',x_3) := U_j\begin{pmatrix} x' \\ hx_3 \end{pmatrix} + U_j^{-1}B_h(x',x_3)\,,$$

where ξ,ζ are smooth functions independent of x_3 to be chosen. We immediately deduce that

(i) $h^{1-\gamma}v_h = v + \dfrac{1}{24}h^2\xi_3 \to v$ in $W^{1,2}(\Omega)$,

(ii) $h^{-\gamma}u_h = u + \dfrac{1}{24}h\xi_{12} \to u$ in $W^{1,2}(\Omega;\mathbb{R}^2)$.

Computing the rescaled gradient, we get

$$\nabla_h y_h = U_j + U_j^{-1}\nabla_h B_h(x',x_3)\,,$$

where

$$\nabla_h B_h = h^\gamma\left[\begin{pmatrix} \nabla u - x_3\nabla^2 v & -h^{-1}\nabla v^T \\ h^{-1}\nabla v & 0 \end{pmatrix} + (x_3\xi+\zeta)\otimes e_3\right] + O(h^{\gamma+1})\,.$$

By construction, we have $\nabla_h y_h = M_h U_j$, where

$$M_h = \mathrm{Id} + U_j^{-1}\nabla_h B_h U_j^{-1}\,.$$